

- 6** The single-celled organism *Thiomargarita namibiensis* measures 700 μm in diameter and is enclosed in a cell wall. The genetic material of *T. namibiensis* is located free in the cytoplasm where it occurs as thousands of copies of circular DNA. Most of the cell is taken up by a large vacuole, which stores essential chemicals for metabolism.

Which statement is correct?

- A** It must be a eukaryote because a diameter of 700 μm is too large for a prokaryote.
- B** It must be a plant because it contains a vacuole.
- C** It must be a plant because it is enclosed in a cell wall.
- D** It must be a prokaryote because its DNA is circular and located in the cytoplasm.